VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Assessment-1

ISWE406P - Agile Development Process and Devops Lab

FACULTY NAME: SWETA BHATTACHARYA

NAME:KARAN SAI GANESH KUMAR

REG NO:23MIS0411

COURSE CODE: ISWE406P

23MIS0411

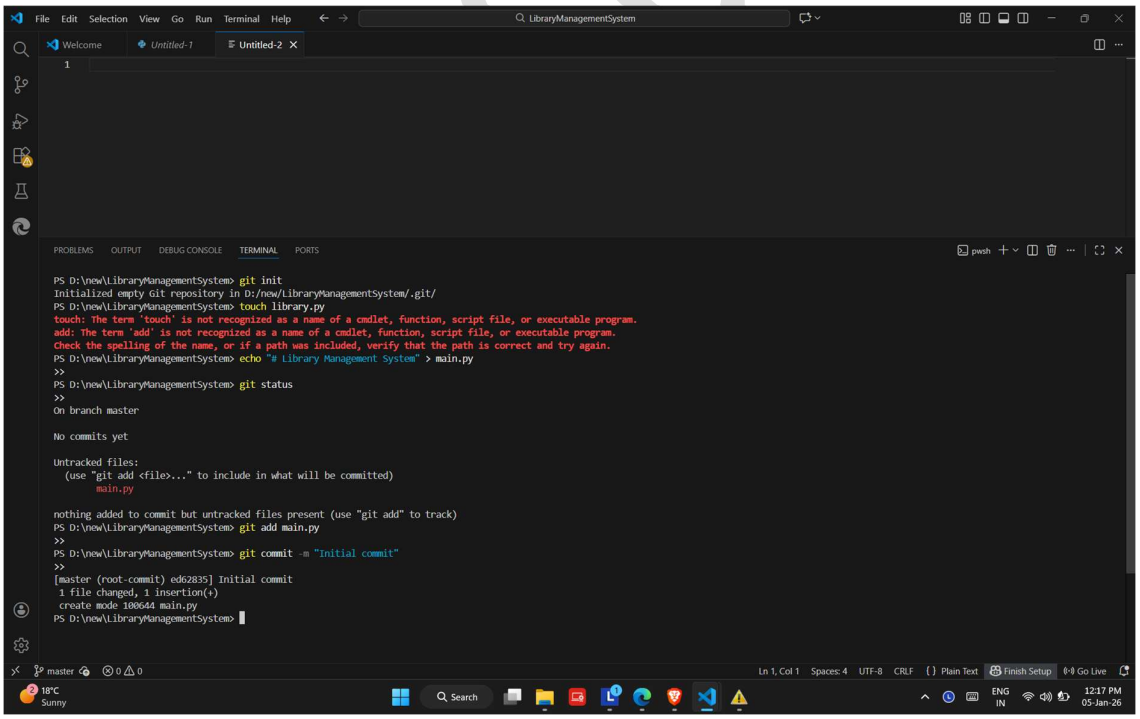
Use appropriate Git Commands to Execute the following case studies in VS Code

Case Study 1: New Project Version Control Setup

Scenario: You are assigned to develop a Library Management System. Before coding, your team decides to enable version control.

Tasks:

- Create a project folder and open it in VS Code
- Initialize a Git repository
- Create a Python file and track it using Git
- Commit the initial version



Case Study 2: Tracking Daily Code Changes

Scenario: You are working on a Student Registration Module. You make small updates daily and

need to record them.

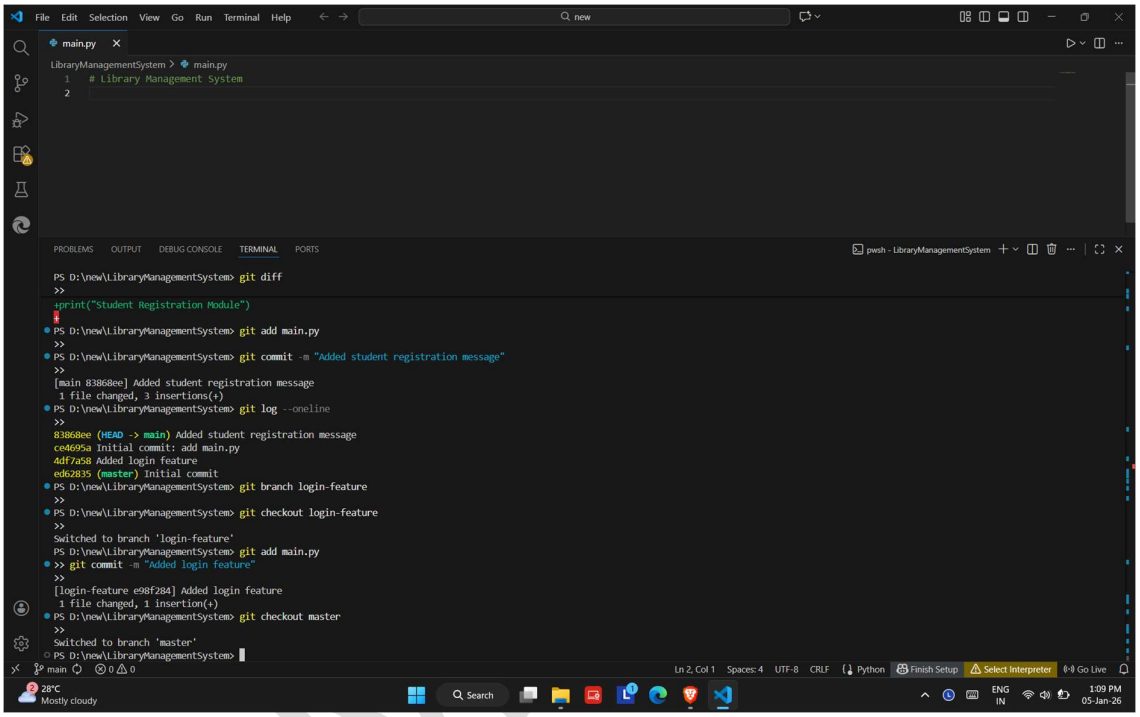
Tasks:

- Modify an existing file

- Check file differences

- Stage and commit changes

- View commit history



Case Study 3: Feature Development Using Branches

Scenario: A new Login feature must be developed without disturbing main code.

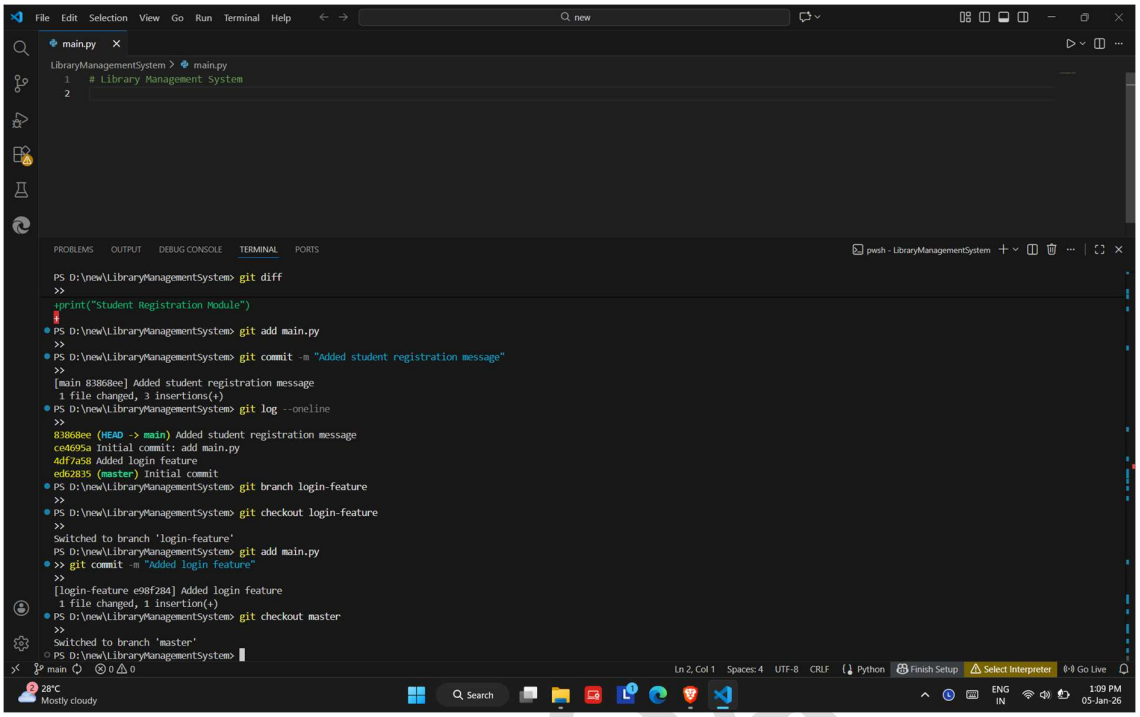
Tasks:

- Create a new branch

- Add login-related code

- Commit changes

- Switch between branches



Case Study 4: Merging Feature into Main Code

Scenario: The login feature is tested and approved.

Tasks:

- Switch to main branch

- Merge feature branch

- Verify merged content

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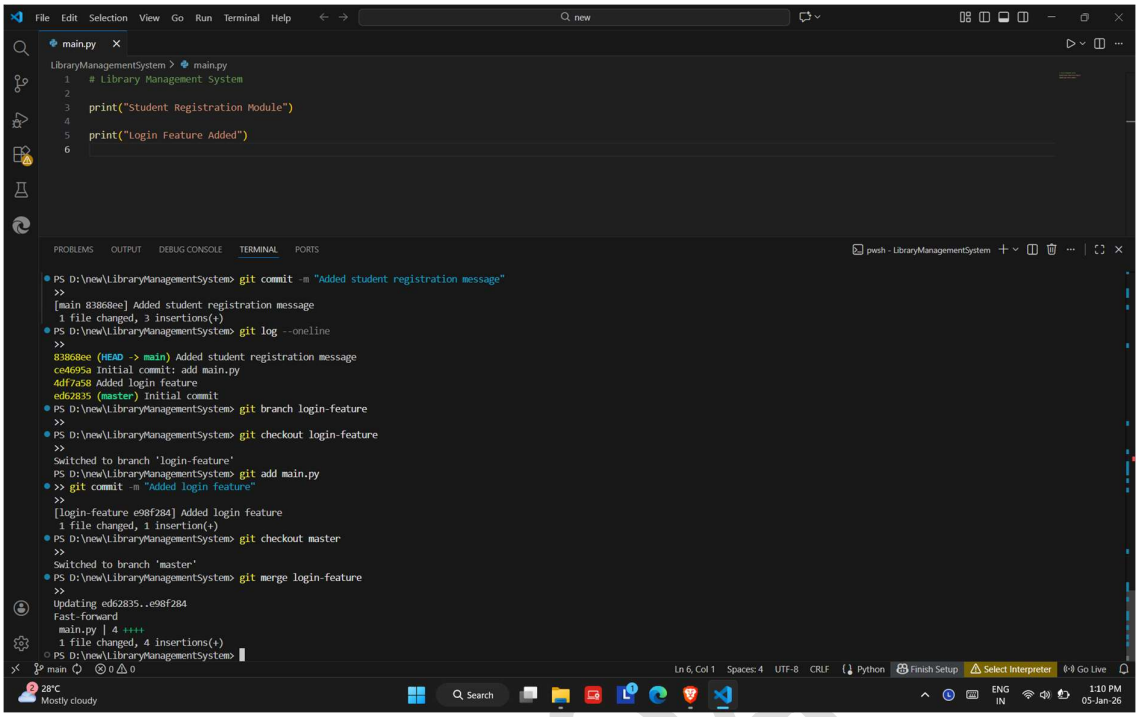

Case Study 5: Working with Remote Repository

Scenario: The project must be shared using GitHub.

Tasks:

- Create remote repository

- Add remote origin

- Push code to GitHub

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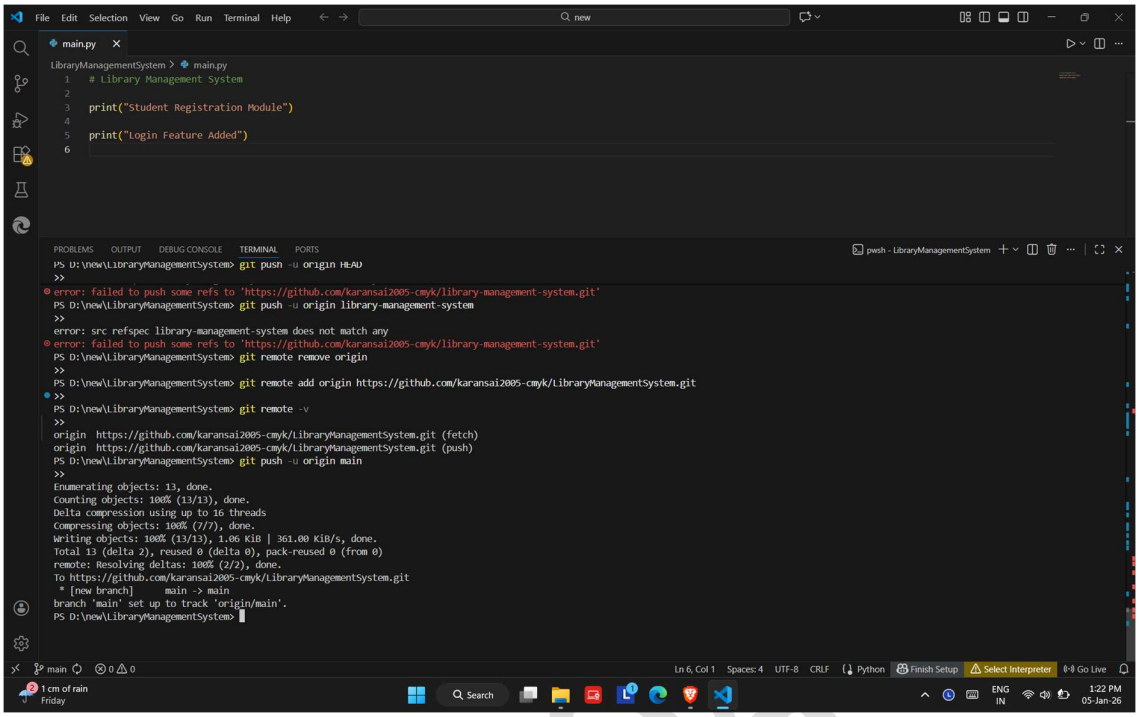


Case Study 6: Undoing Mistakes

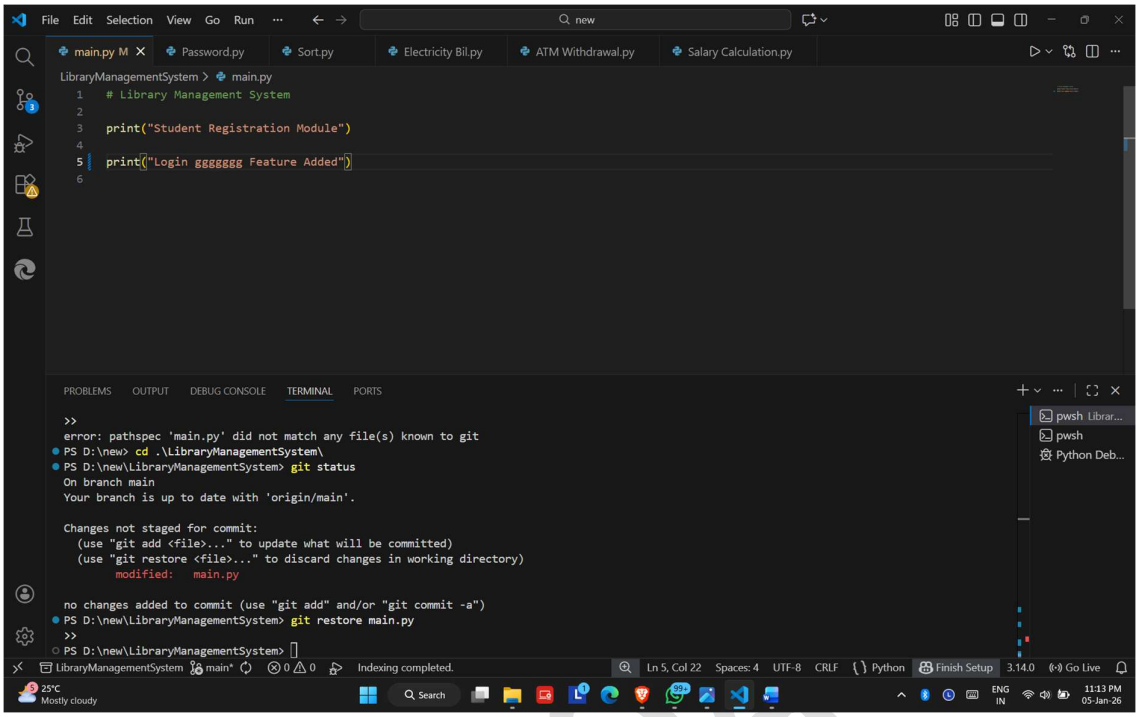
Scenario: A file is modified incorrectly.

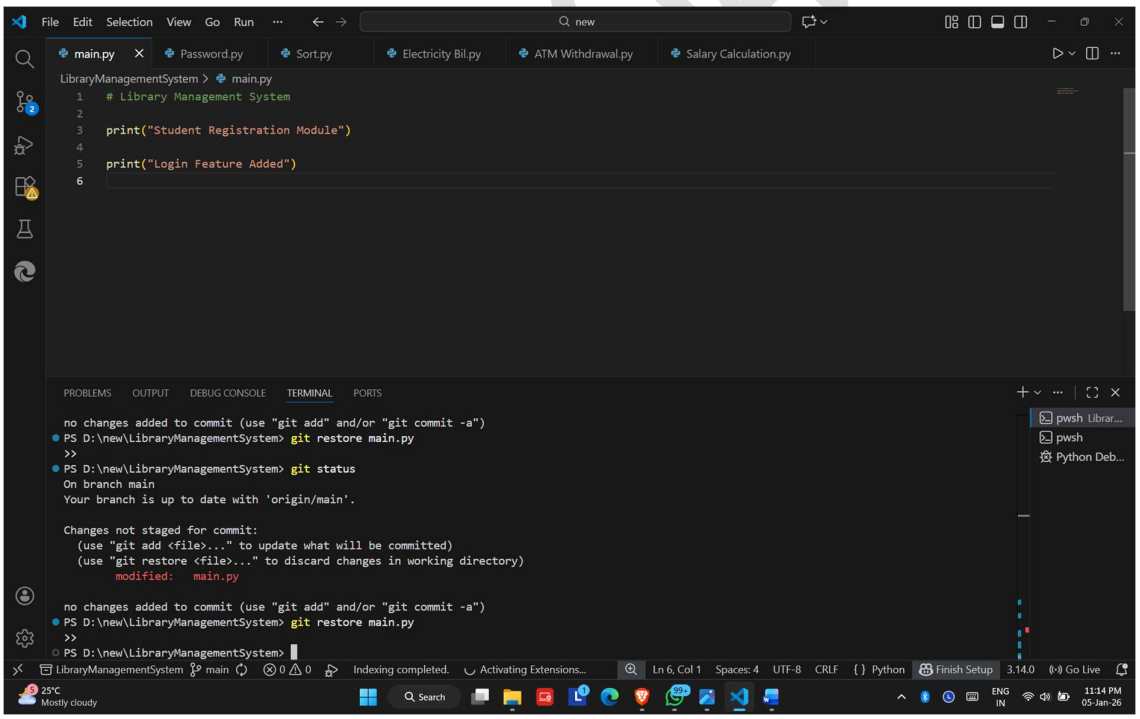
Tasks:

- View file changes

- Discard uncommitted changes

23MISO4





Case Study 7: Handling Merge Conflicts

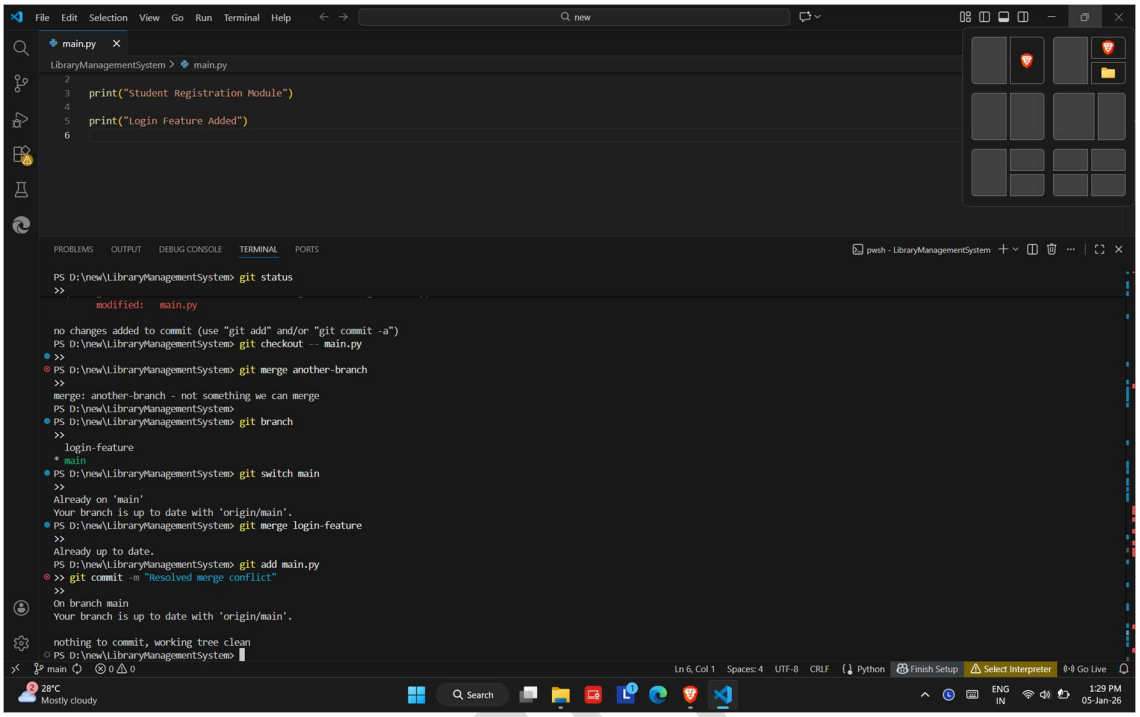
Scenario: Two developers edited the same file.

Tasks:

- Merge branches

- Resolve conflicts

- Commit resolved changes



Case Study 8: Ignoring Unwanted Files

Scenario: Log and temporary files should not be tracked.

Tasks:

- Create .gitignore file

- Add ignore patterns

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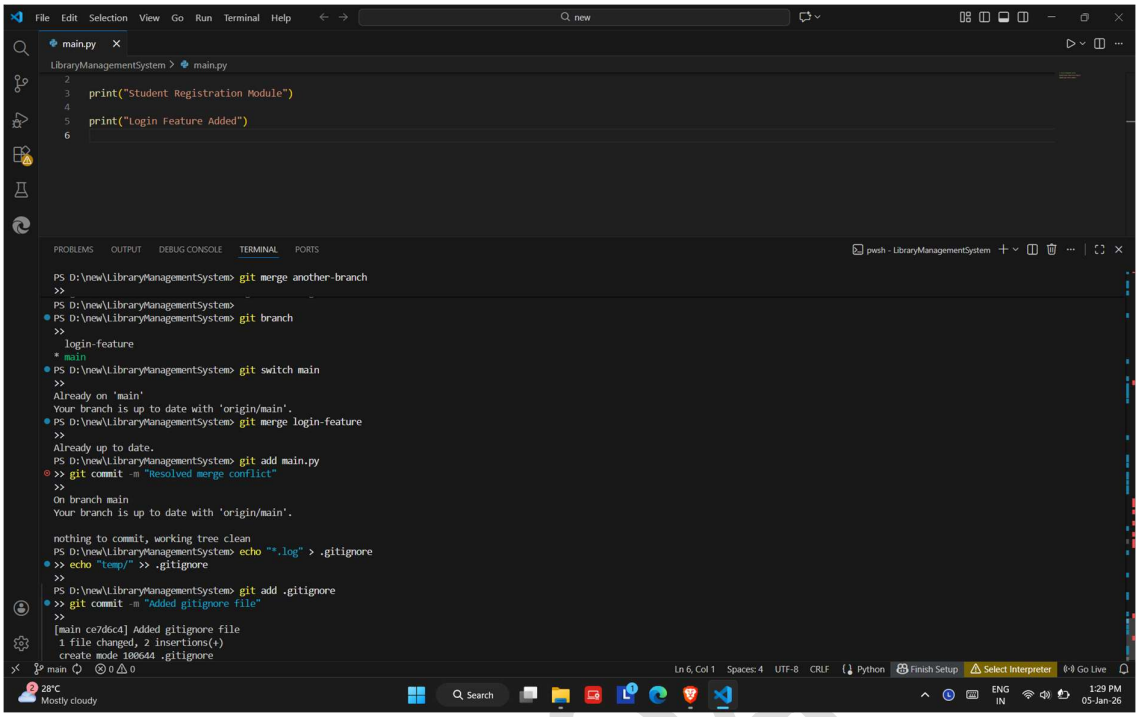

Case Study 9: Collaborating with Team

Scenario: A teammate pushed new changes.

Tasks:

- Fetch latest changes

- Pull updates

- Resolve conflicts if any

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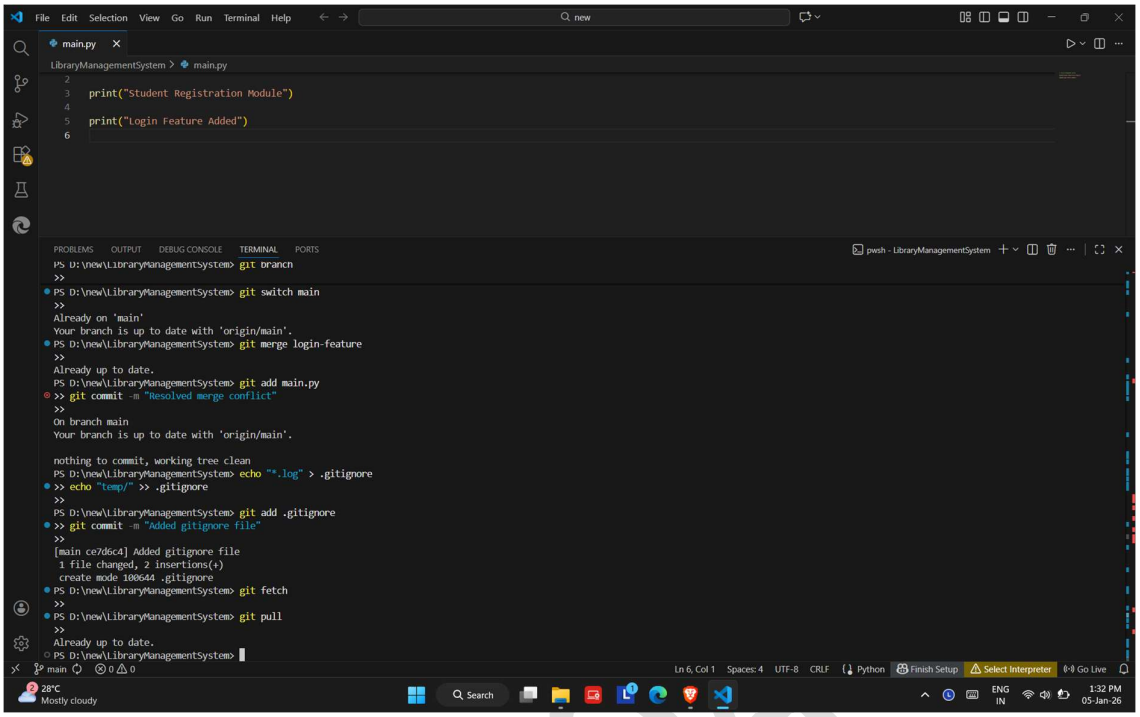

Case Study 10: Final Project Release

Scenario:

Project is ready for final submission.

Tasks:

- Review commit history

- Push final code

- Prepare release version

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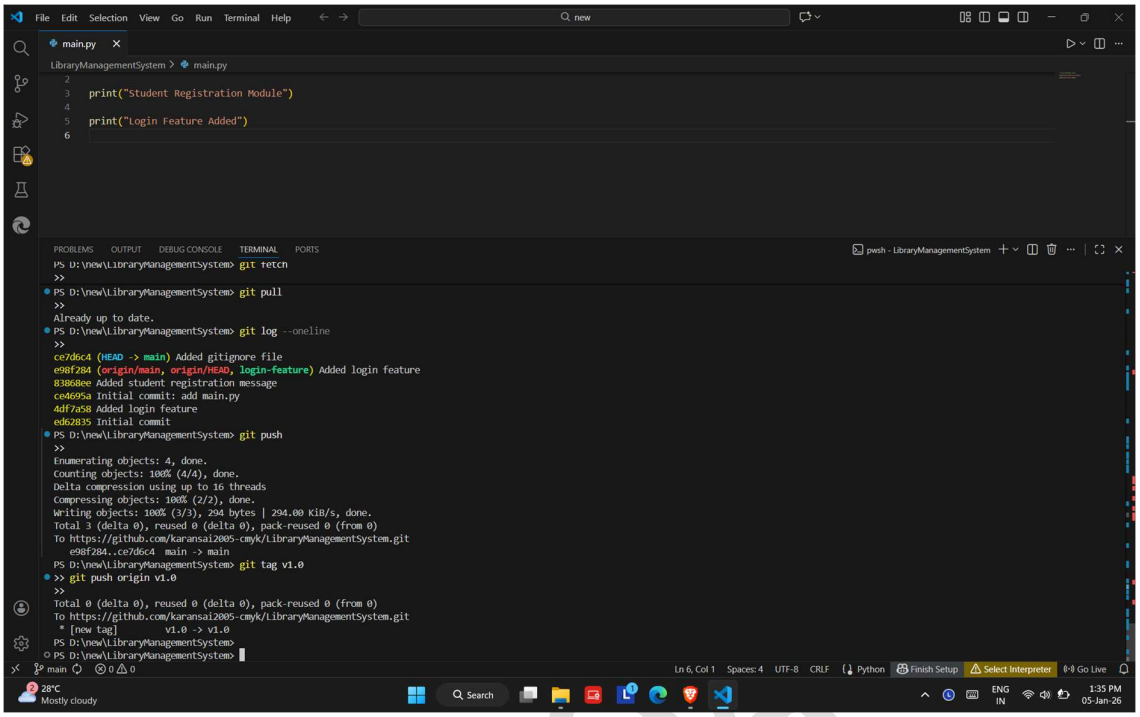
File Edit Selection View Go Run Terminal Help
new
main.py
LibraryManagementSystem > main.py
print("Student Registration Module")
print("Login Feature Added")
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
pwsh - LibraryManagementSystem
PS D:\new\LibraryManagementSystem> git fetch
>>
PS D:\new\LibraryManagementSystem> git pull
>>
Already up to date.
PS D:\new\LibraryManagementSystem> git log --oneline
>>
ce7d6c4 (HEAD -> main) Added gitignore file
e98f284 (origin/main, origin/HEAD, login-feature) Added login feature
83868ee Added student registration message
c04695a Initial commit: add main.py
4df7a58 Added login feature
ed62835 Initial commit
PS D:\new\LibraryManagementSystem> git push
>>
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 294 bytes | 294.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/karansai2005-cmyk/LibraryManagementSystem.git
e98f284..ce7d6c4 main -> main
PS D:\new\LibraryManagementSystem> git tag v1.0
>> git push origin v1.0
>>
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/karansai2005-cmyk/LibraryManagementSystem.git
* [new tag] v1.0 -> v1.0
PS D:\new\LibraryManagementSystem>
PS D:\new\LibraryManagementSystem>
main
Ln 6, Col 1 Spaces: 4 UTF-8 CRLF Python Finish Setup Select Interpreter Go Live
28°C
Mostly cloudy
Search
ENG
IN
1:35 PM
05-Jan-26

23MIS04

Python programs

Question 1.

A website requires the users to input username and password to register. Write a program to check the validity of password input by users.

Following are the criteria for checking the password:

1. At least 1 letter between [a-z]

2. At least 1 number between [0-9]

1. At least 1 letter between [A-Z]

3. At least 1 character from [$#@]

4. Minimum length of transaction password: 6

5. Maximum length of transaction password: 12

Your program should accept a sequence of comma separated passwords and will check them according to the above criteria. Passwords that match the criteria are to be printed, each separated by a comma.

Example

If the following passwords are given as input to the program:

ABd1234@1,a F1#,2w3E*,2We3345

Then, the output of the program should be:

ABd1234@1

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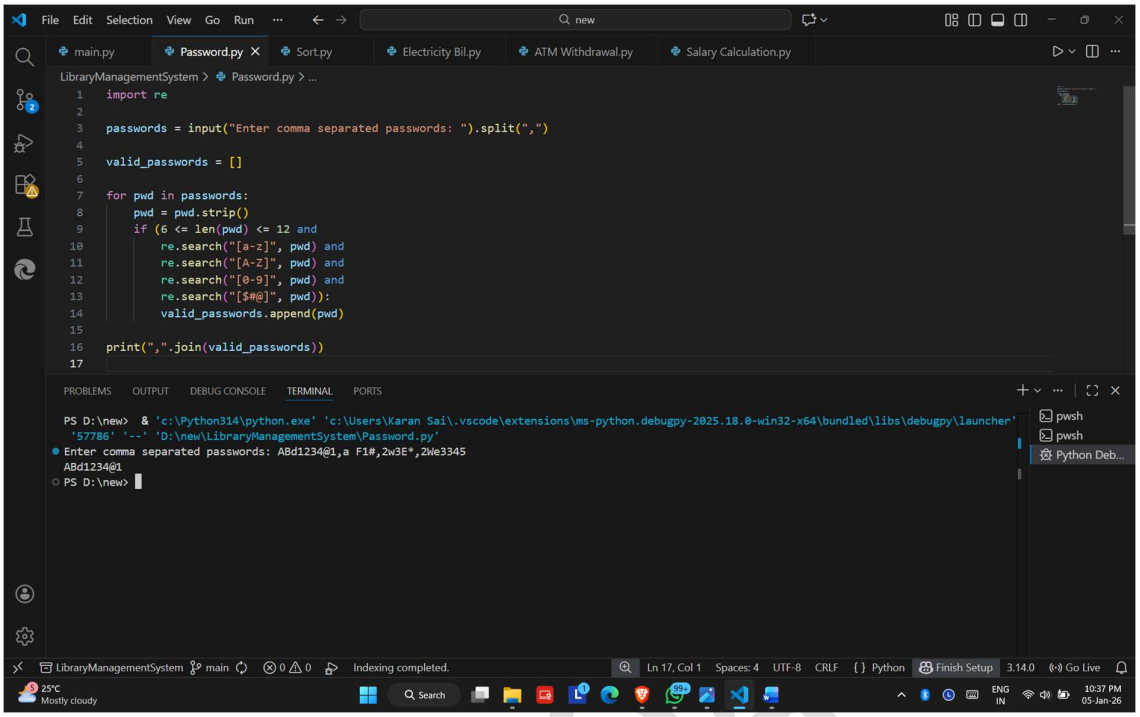
File Edit Selection View Go Run
new
main.py
Password.py
Sort.py
Electricity Bil.py
ATM Withdrawal.py
Salary Calculation.py
LibraryManagementSystem > Password.py > ...
import re
passwords = input("Enter comma separated passwords: ").split(",")
valid_passwords = []
for pwd in passwords:
pwd = pwd.strip()
if (6 <= len(pwd) <= 12 and
re.search("[a-z]", pwd) and
re.search("[A-Z]", pwd) and
re.search("[0-9]", pwd) and
re.search("[$#@]", pwd)):
valid_passwords.append(pwd)
print(",".join(valid_passwords))
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\new> & 'c:\Python314\python.exe' 'c:\Users\Karan Sai\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57786' '--' 'D:\new\LibraryManagementSystem\Password.py'
Enter comma separated passwords: ABd1234@1,a F1#,2w3E*,2We3345
ABd1234@1
PS D:\new>
pwsh
pwsh
Python Deb...
LibraryManagementSystem main
Indexing completed.
Ln 17, Col 1 Spaces: 4 UTF-8 CRLF {} Python Finish Setup 3.14.0 Go Live
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ENG
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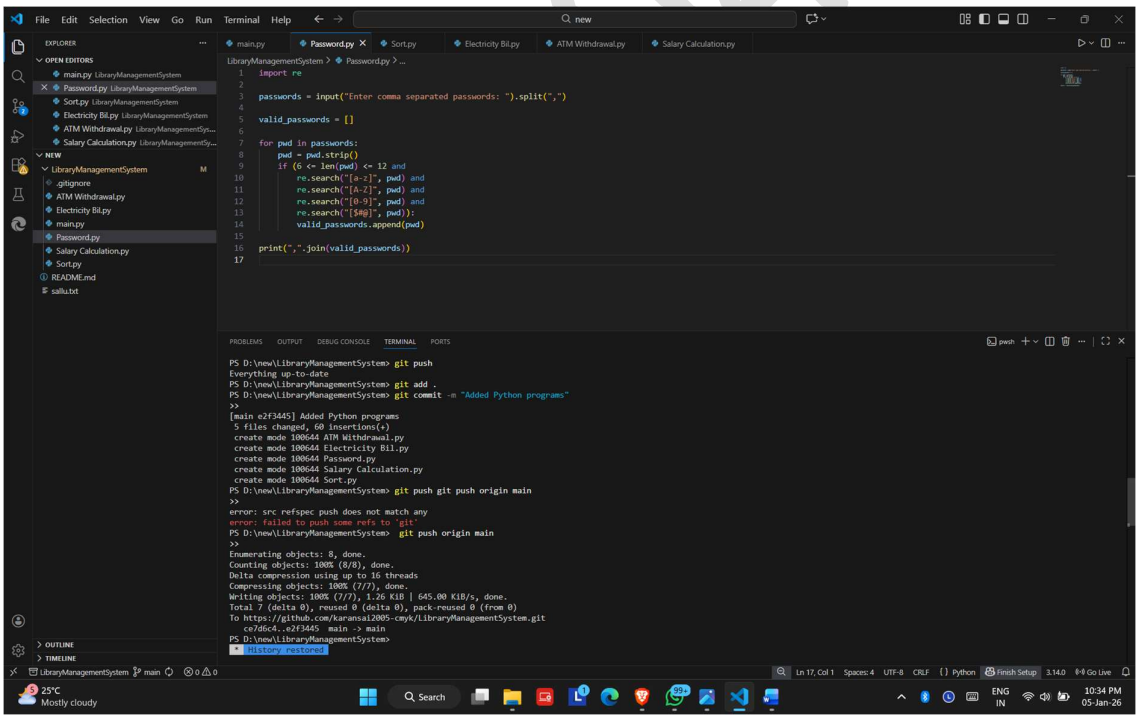
File Edit Selection View Go Run Terminal Help
new
EXPLORER
OPEN EDITORS
main.py LibraryManagementSystem
Password.py LibraryManagementSystem
Sort.py LibraryManagementSystem
Electricity Bil.py LibraryManagementSystem
ATM Withdrawal.py LibraryManagementSys...
Salary Calculation.py LibraryManagementSy...
NEW
LibraryManagementSystem M
.gitignore
ATM Withdrawal.py
Electricity Bil.py
main.py
Password.py
Salary Calculation.py
Sort.py
README.md
sallu.txt
OUTLINE
TIMELINE
LibraryManagementSystem > Password.py > ...
import re
passwords = input("Enter comma separated passwords: ").split(",")
valid_passwords = []
for pwd in passwords:
pwd = pwd.strip()
if (6 <= len(pwd) <= 12 and
re.search("[a-z]", pwd) and
re.search("[A-Z]", pwd) and
re.search("[0-9]", pwd) and
re.search("[$#@]", pwd)):
valid_passwords.append(pwd)
print(",".join(valid_passwords))
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\new\LibraryManagementSystem> git push
Everything up-to-date
PS D:\new\LibraryManagementSystem> git add .
PS D:\new\LibraryManagementSystem> git commit -m "Added Python programs"
>>
[main e2f3445] Added Python programs
5 files changed, 60 insertions(+)
create mode 100644 ATM Withdrawal.py
create mode 100644 Electricity Bil.py
create mode 100644 Password.py
create mode 100644 Salary Calculation.py
create mode 100644 Sort.py
PS D:\new\LibraryManagementSystem> git push git push origin main
>>
error: src refspec push does not match any
error: failed to push some refs to 'git'
PS D:\new\LibraryManagementSystem> git push origin main
>>
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.26 KiB | 645.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/karansai2005-cmyk/LibraryManagementSystem.git
ce7d6c4..e2f3445 main -> main
PS D:\new\LibraryManagementSystem>
History restored
LibraryManagementSystem main
Ln 17, Col 1 Spaces: 4 UTF-8 CRLF {} Python Finish Setup 3.14.0 Go Live
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05-Jan-26

Question 2.

You are required to write a program to sort the (name, age, height) tuples by ascending order where name is string, age and height are numbers. The tuples are input by console. The sort criteria is:

1: Sort based on name;

2: Then sort based on age;

3: Then sort by score.

The priority is that name > age > score.

If the following tuples are given as input to the program:

Tom,19,80

John,20,90

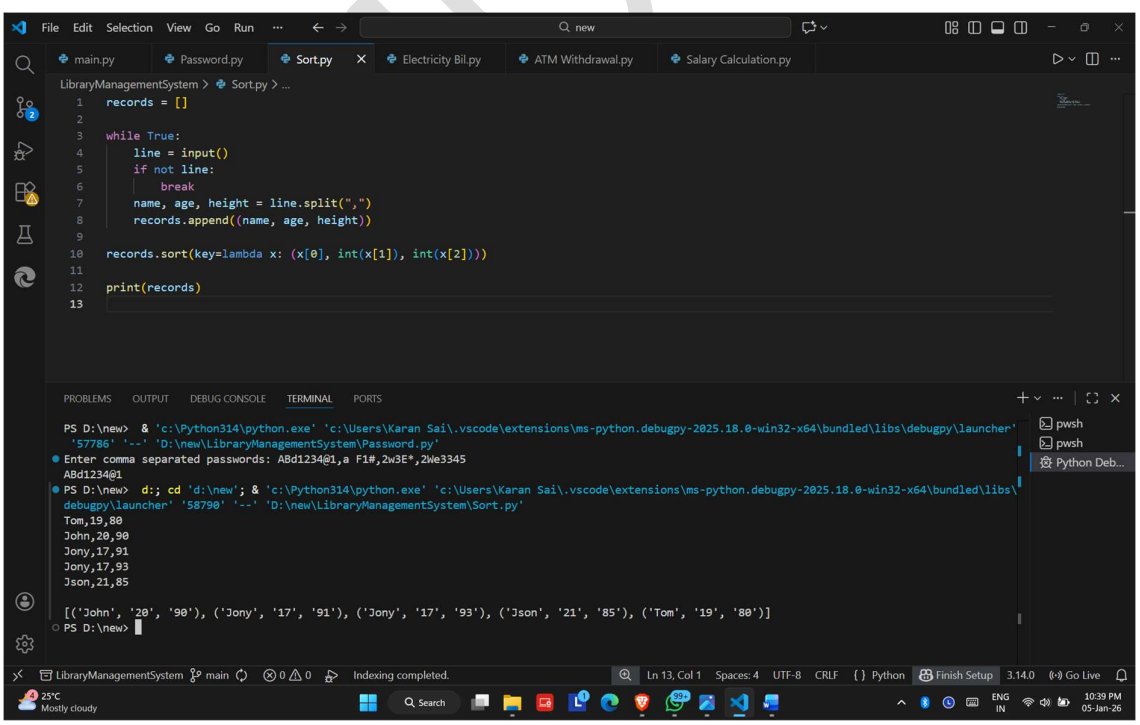
Jony,17,91

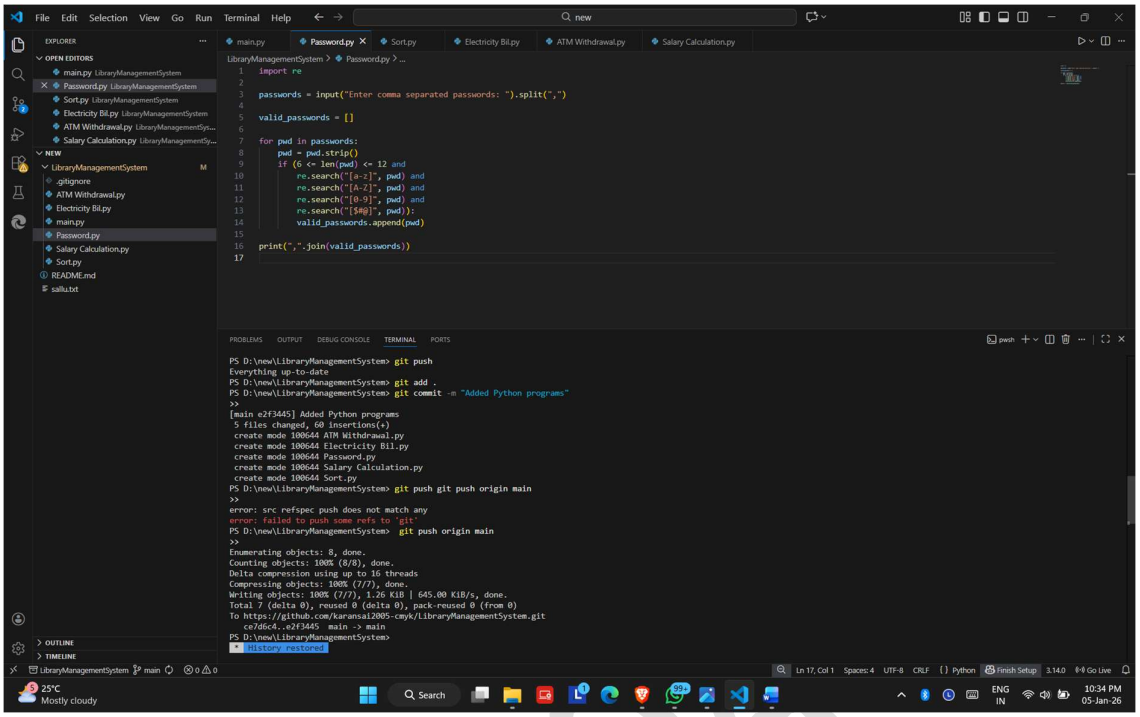
Jony,17,93

Json,21,85

Then, the output of the program should be:

[('John', '20', '90'), ('Jony', '17', '91'), ('Jony', '17', '93'), ('Json', '21', '85'), ('Tom', '19', '80')]





Question 3:

An electricity board calculates charges based on usage.

Write a program that:

• Takes number of units consumed

• Calculates bill:

o First 100 units → ₹2/unit

o Next 100 units → ₹3/unit

o Remaining → ₹5/unit

• Displays total bill

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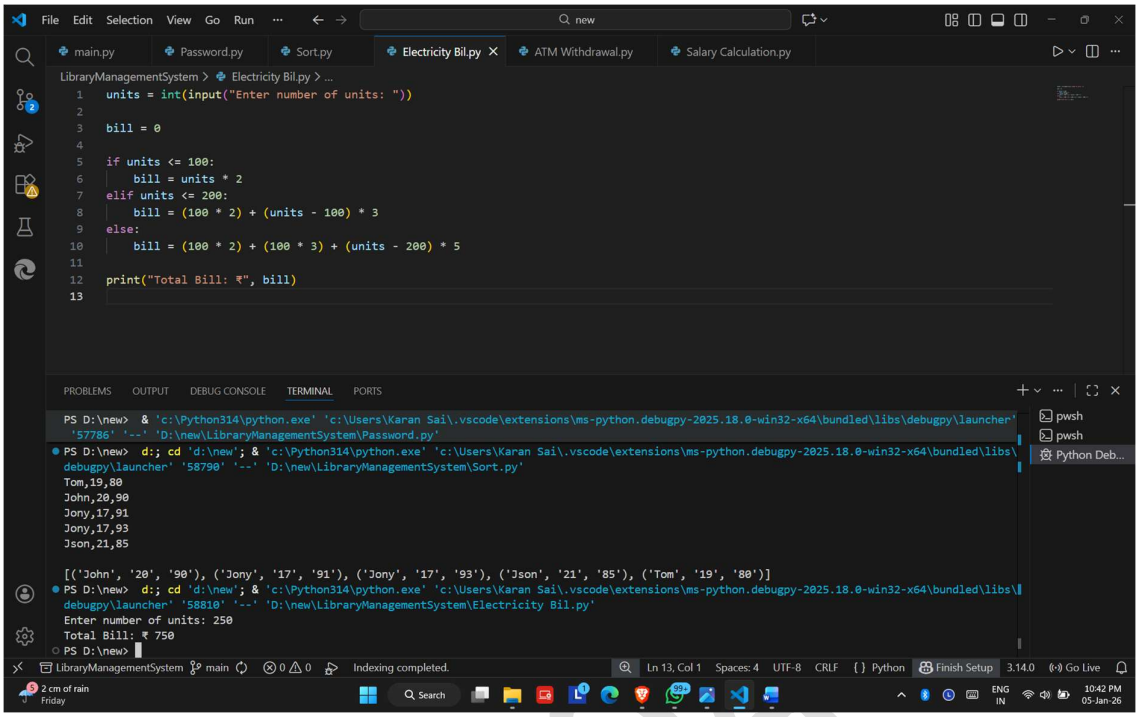
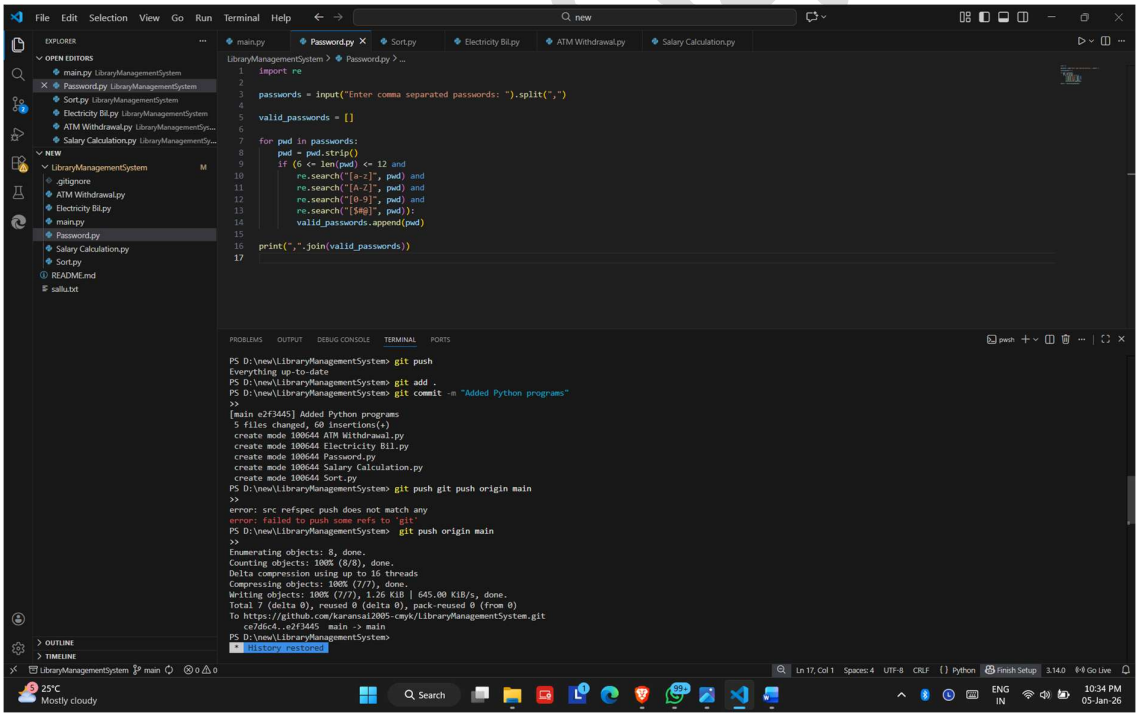

File Edit Selection View Go Run
new
main.py
Password.py
Sort.py
Electricity Bil.py
ATM Withdrawal.py
Salary Calculation.py
LibraryManagementSystem > Electricity Bil.py > ...
units = int(input("Enter number of units: "))
bill = 0
if units <= 100:
bill = units * 2
elif units <= 200:
bill = (100 * 2) + (units - 100) * 3
else:
bill = (100 * 2) + (100 * 3) + (units - 200) * 5
print("Total Bill: ₹", bill)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\new> & 'c:\Python314\python.exe' 'c:\Users\Karan Sai\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57786' '--' 'D:\new\LibraryManagementSystem\Password.py'
PS D:\new> d:; cd 'd:\new'; & 'c:\Python314\python.exe' 'c:\Users\Karan Sai\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58790' '--' 'D:\new\LibraryManagementSystem\Sort.py'
Tom,19,80
John,20,90
Jony,17,91
Jony,17,93
Json,21,85
[('John', '20', '90'), ('Jony', '17', '91'), ('Jony', '17', '93'), ('Json', '21', '85'), ('Tom', '19', '80')]
PS D:\new> d:; cd 'd:\new'; & 'c:\Python314\python.exe' 'c:\Users\Karan Sai\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58810' '--' 'D:\new\LibraryManagementSystem\Electricity Bil.py'
Enter number of units: 250
Total Bill: ₹ 750
PS D:\new>
pwsh
pwsh
Python Deb...
LibraryManagementSystem main
Indexing completed.
Ln 13, Col 1 Spaces: 4 UTF-8 CRLF {} Python Finish Setup 3.14.0 Go Live
2 cm of rain
Friday
Search
ENG IN
10:42 PM
05-Jan-26


File Edit Selection View Go Run Terminal Help
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EXPLORER
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main.py LibraryManagementSystem
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ATM Withdrawal.py LibraryManagementSys...
Salary Calculation.py LibraryManagementSy...
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Electricity Bil.py
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Salary Calculation.py
Sort.py
README.md
sallu.txt
LibraryManagementSystem > Password.py > ...
import re
passwords = input("Enter comma separated passwords: ").split(",")
valid_passwords = []
for pwd in passwords:
pwd = pwd.strip()
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re.search("[a-z]", pwd) and
re.search("[A-Z]", pwd) and
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valid_passwords.append(pwd)
print(",".join(valid_passwords))
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\new\LibraryManagementSystem> git push
Everything up-to-date
PS D:\new\LibraryManagementSystem> git add .
PS D:\new\LibraryManagementSystem> git commit -m "Added Python programs"
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create mode 100644 ATM Withdrawal.py
create mode 100644 Electricity Bil.py
create mode 100644 Password.py
create mode 100644 Salary Calculation.py
create mode 100644 Sort.py
PS D:\new\LibraryManagementSystem> git push git push origin main
>>
error: src refspec push does not match any
error: failed to push some refs to 'git'
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>>
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.26 KiB | 645.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/karansai2005-cmyk/LibraryManagementSystem.git
ce7d6c4..e2f3445 main -> main
PS D:\new\LibraryManagementSystem>
History restored
OUTLINE
TIMELINE
LibraryManagementSystem main
Ln 17, Col 1 Spaces: 4 UTF-8 CRLF {} Python Finish Setup 3.14.0 Go Live
25°C
Mostly cloudy
Search
ENG IN
10:34 PM
05-Jan-26

Question 4:

An ATM machine allows withdrawals only if sufficient balance is available.

Write a program that:

• Stores a user's balance

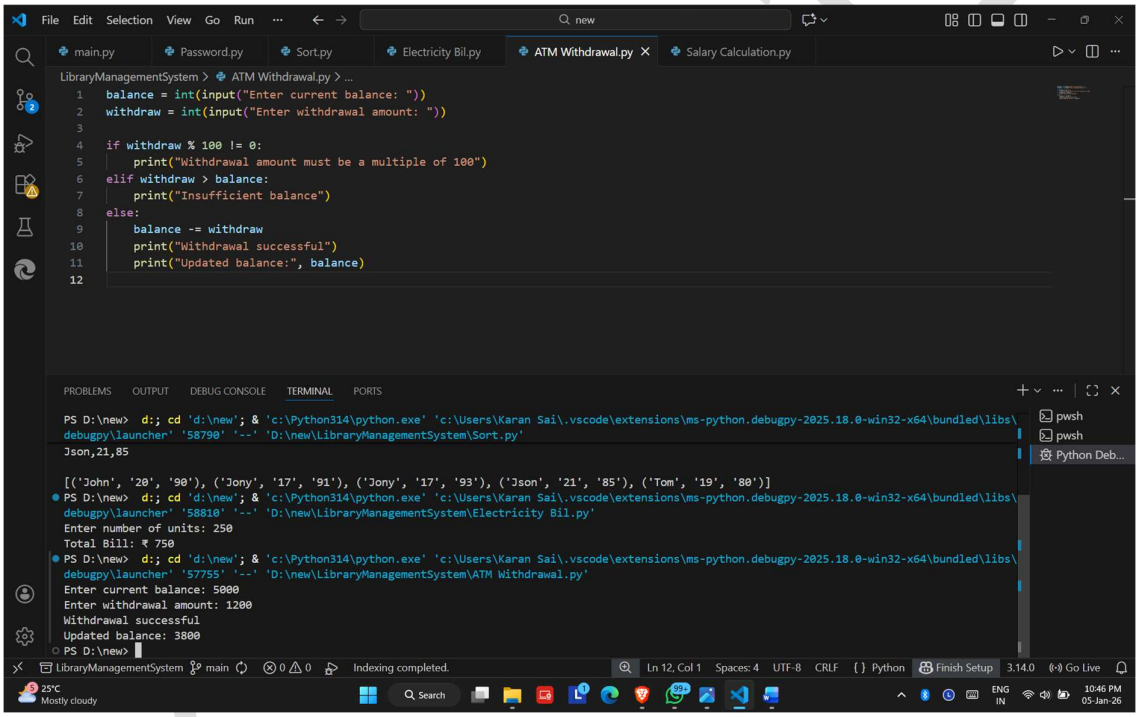
• Takes withdrawal amount as input

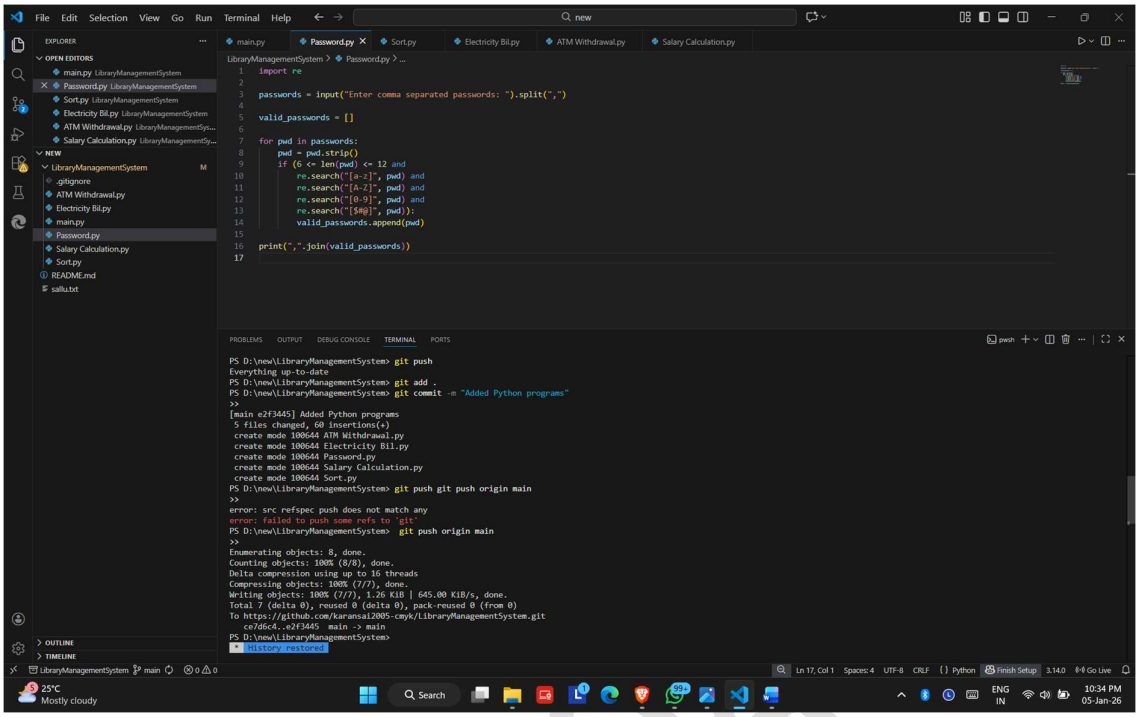
• Checks:

o Amount must be a multiple of 100

o Balance must be sufficient

• Deducts amount and displays updated balance





Question 5:

A company calculates monthly salary based on basic pay.

Write a program that:

• Accepts basic salary

• Calculates:

o HRA = 20% of basic

o DA = 10% of basic

o Tax = 5% of total salary

• Displays net salary

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```python
basic = float(input("Enter basic salary: "))

hra = basic * 0.20
da = basic * 0.10
gross_salary = basic + hra + da
tax = gross_salary * 0.05
net_salary = gross_salary - tax

print("Net Salary:", net_salary)
```

```
PS D:\new>  d:; cd 'd:\new'; & 'c:\Python314\python.exe' 'c:\Users\Karan Sai\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58790' '--' 'D:\new\LibraryManagementSystem\Sort.py'
PS D:\new>  d:; cd 'd:\new'; & 'c:\Python314\python.exe' 'c:\Users\Karan Sai\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '58810' '--' 'D:\new\LibraryManagementSystem\Electricity Bil.py'
Enter number of units: 250
Total Bill: ₹ 750
PS D:\new>  d:; cd 'd:\new'; & 'c:\Python314\python.exe' 'c:\Users\Karan Sai\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '57755' '--' 'D:\new\LibraryManagementSystem\ATM Withdrawal.py'
Enter current balance: 5000
Enter withdrawal amount: 1200
Withdrawal successful
Updated balance: 3800
PS D:\new>  d:; cd 'd:\new'; & 'c:\Python314\python.exe' 'c:\Users\Karan Sai\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '53057' '--' 'D:\new\LibraryManagementSystem\Salary Calculation.py'
Enter basic salary: 20000
Net Salary: 24700.0
PS D:\new>
```

```python
import re

passwords = input("Enter comma separated passwords: ").split(",")

valid_passwords = []

for pwd in passwords:
    pwd = pwd.strip()
    if (6 <= len(pwd) <= 12 and
        re.search("[a-z]", pwd) and
        re.search("[A-Z]", pwd) and
        re.search("[0-9]", pwd) and
        re.search("[$#@]", pwd)):
        valid_passwords.append(pwd)

print(",".join(valid_passwords))
```

```
PS D:\new\LibraryManagementSystem> git push
Everything up-to-date
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PS D:\new\LibraryManagementSystem> git commit -m "Added Python programs"
>>
[main e2f3445] Added Python programs
 5 files changed, 60 insertions(+)
 create mode 100644 ATM Withdrawal.py
 create mode 100644 Electricity Bil.py
 create mode 100644 Password.py
 create mode 100644 Salary Calculation.py
 create mode 100644 Sort.py
PS D:\new\LibraryManagementSystem> git push git push origin main
>>
error: src refspec push does not match any
error: failed to push some refs to 'git'
PS D:\new\LibraryManagementSystem>  git push origin main
>>
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.26 KiB | 645.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/karansai2005-cmyk/LibraryManagementSystem.git
   ce7d6c4..e2f3445  main -> main
PS D:\new\LibraryManagementSystem>
History restored
```

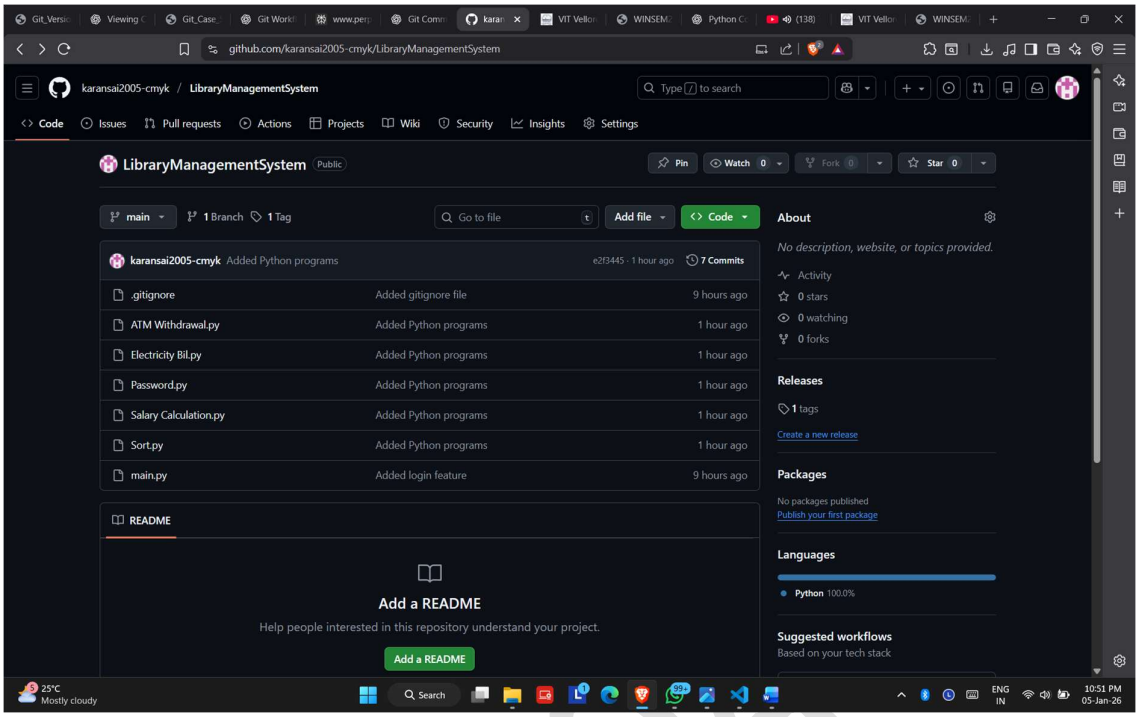
github.com/karansai2005-cmyk/LibraryManagementSystem
karansai2005-cmyk / LibraryManagementSystem
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
LibraryManagementSystem
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
1 Tag
Go to file
Add file
Code
karansai2005-cmyk Added Python programs
e2f3445 · 1 hour ago
7 Commits
.gitignore
Added gitignore file
9 hours ago
ATM Withdrawal.py
Added Python programs
1 hour ago
Electricity Bil.py
Added Python programs
1 hour ago
Password.py
Added Python programs
1 hour ago
Salary Calculation.py
Added Python programs
1 hour ago
Sort.py
Added Python programs
1 hour ago
main.py
Added login feature
9 hours ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
1 tags
Create a new release
Packages
No packages published
Publish your first package
Languages
Python 100.0%
Suggested workflows
Based on your tech stack
25°C
Mostly cloudy
Search
ENG
IN
10:51 PM
05-Jan-26

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